

Task 5:

Break first:

NISQA inference. No question. The model loads fresh on every upload right now. That works for a demo with test files, but 5,000 workers over 48 hours means hundreds of concurrent requests all initializing the same model. CPU max utilize, RAM spikes, and a big problem.

SQLite DB issue. When 50 workers submit at once, 49 get "database is locked." It doesn't fail, it hangs, because mainly it design for edge devices not for multiple users performing operation as same time.

What I'd Change Before Launch

Load models once at startup. Keep SQUIM (this already implemented) and NISQA in memory. That alone drops per-request latency and fix the ram problem.

Add a background job queue. Don't analyze inside the HTTP request. User uploads, gets an immediate "received" confirmation with a job ID, and background job running until finished. And create API (FastAPI) on top of this, this can easily handle background job.

Swap SQLite for PostgreSQL. Postgres handles multiple operations at once without getting lags, and gives real-time status.

Deduplicate before analyzing. If a worker clicks submit twice, or two workers share the same recording, skip the ML pipeline and return cached results. Saves compute and money. It can implemented by transcript.

Add limit. Max file size (10MB), max duration (5 min), format whitelist (WAV, MP3 only). One 500MB upload will throw error message.

Limit GPU workers. Use a fixed-size pool (4 concurrent inference jobs max). Everything else queues. GPU memory is finite and expensive; don't let 100 simultaneous analyses fight for limited VRAM.

Add rate limiting. 5 uploads per minute per phone number. Prevents abuse and protects the queue from being flooded by a script or a confused user hammering submit.

Monitor everything. Upload success rate, queue depth, average inference time, GPU utilization, storage growth and error rate by stage.

Integrate LLM agent. LLM multimodal or audio model (can utilize text model too using transcript) take audio as input and do different kind of work like authentication, auto tag, categorize, give transcript (or summary), execute workflow and others types of operations.